

2(a)	allow mosquito for <u>Anopheles</u> throughout allow pathogen for <u>Plasmodium</u> throughout any three from role of <i>Anopheles</i> in transmission cycle ; e.g. <i>Anopheles</i> is, a vector of <i>Plasmodium</i> / <i>Anopheles</i> passes <i>Plasmodium</i> from infected person to uninfected person <i>insecticide on nets and on surfaces</i> kills <i>Anopheles</i> before it can take <u>blood</u> from an infected person ; kills <i>Anopheles</i> before it can transfer <u>blood</u> with <i>Plasmodium</i> to uninfected person ; <i>presence of nets</i> protect people, when sleeping / at time when <i>Anopheles</i> is, active / feeding ; <i>general</i> prevent <i>Plasmodium</i> from completing its life cycle ; AW AVP ; <i>idea of</i> reducing population size of mosquitoes use of different insecticides on net and IRS to avoid insecticide resistance	3
2(b)(i)	any one from increase in the use of ITN over time ; decrease in the use of IRS only ; proportion of population protected by ITN only has increased ;	1
2(b)(ii)	any one from increase is not steep enough, to make a large enough difference / so disease will still be able to spread ; a large proportion of the population is still at risk, so people will still contract the disease ; A figures from Fig. 2.1 WHO targets may not be met, so hindering progress in the fight against malaria ; AW AVP ; e.g. suggests that the, provision of / distribution of / access to, ITN is not adequate ;	1

Question	Answer	Marks
2(b)(iii)	<i>any one suggestion from</i> ITN more effective in control than IRS ; more cost effective to provide ITN ; lack of personnel to carry out work required for IRS ; <i>ref. to</i> insecticide on net may be more effective at killing ; AVP ; e.g. outside agencies / AW, provide ITN but do not help with IRS	1
2(c)	<i>any three from</i> higher concentration of antibodies ; faster production of antibodies ; because of presence of memory, B-lymphocytes / B cells ; higher numbers of specific B-lymphocytes, so increased chance of faster recognition / because of clonal expansion in first response ; AVP ; e.g. also more memory T-cells to stimulate B-lymphocyte response <i>ref. to</i> higher concentration antibodies in circulation remaining after recovery	3
2(d)	<i>any two from</i> result of an autoimmune disease / AW ; antibodies produced against, self-antigens / antigens on body cells <i>or</i> antibodies bind to self-antigens / antigens on (own) body cells ; detail ; e.g. prevents functioning of muscle cells binds to receptors on muscle cells	2

Question	Answer	Marks
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